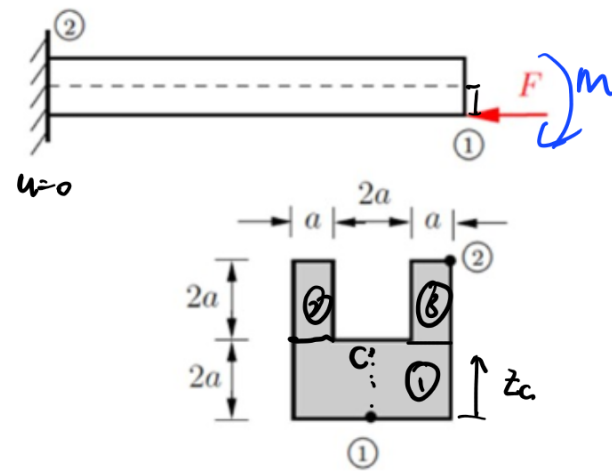


Homework 4.9 A cantilever beam with the sketched cross section is loaded by the force F at point (1). Determine the normal stresses at point (2) at the support.



$$z_c = \frac{\sum A_i z_i}{\sum A_i} = \frac{2a(4a) \cdot a + 2(2a)(a) \cdot (3a)}{2a(a)(2a) + 2a(4a)} = \frac{20a^3}{12a^3} = \frac{5}{3}a$$

$$I_y = \left[\frac{1}{12} (4a)(2a)^3 + \left(\frac{2}{3}a \right)^2 (8a^2) \right] + 2 \left[\frac{1}{12} (a)(2a)^3 + \left(\frac{7}{3}a \right)^2 (2a^2) \right]$$

$$= \frac{8}{3}a^4 + \frac{32}{9}a^4 + \frac{8}{6}a^4 + \frac{64}{9}a^4$$

$$= \frac{44}{3}a^4$$

$$N = -F \quad \sigma_N = -\frac{F}{12a^2} \quad \sigma_B = \frac{M}{I} z = -\frac{5}{3}aFz \times \frac{1}{\frac{44}{3}a^4} = -\frac{5}{44} \frac{Fz}{a^3}$$

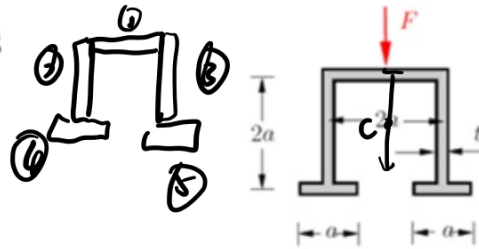
$$M = -Fz_c(P)$$

At point (2)

$$\sigma = \sigma_N + \sigma_B(z_c) = -\frac{F}{12a^2} + \frac{5}{44} \frac{Fz}{a^3} \left(+\frac{7}{3}a \right) = -\frac{F}{12a^2} + \frac{35}{132a^2} = \frac{2}{11} \frac{F}{a^2}$$

Homework 4.10 A cantilever beam with the depicted cross section (constant wall thickness t , $t \ll a$ is subjected to a concentrated force F at one end.

Determine the maximum stress in the cross section at the support.



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$$z_c = \frac{\sum A_i z_i}{\sum A_i} = \frac{2at(\frac{a}{2}) + 2(2at)(a + \frac{t}{2}) + 2At(2a + \frac{t}{2})}{2at + 2(2at) + 2at} = \frac{at^2 + 4a^2t + 2at^2 + 4a^2t + at^2}{8at}$$

$\therefore t \ll a$, we ignore higher order terms of t

$$\therefore z_c = \frac{8a^2t}{8at} = a$$

$$I_y = \frac{1}{12}(2a)(t)^3 + (a)^2(2at) + 2[(\frac{1}{12})(t)(2a)^3] + 2[(\frac{1}{12})(a)(t)^3 + (a)^2(at)]$$

$$= 2a^3t + \frac{8}{3}a^3t + 2a^3t = \frac{16}{3}a^3t$$

section modulus $W = \frac{I_y}{z_{\max}}$

$z_{\max} = a$

$$W = \frac{\frac{16}{3}a^3t}{a} = \frac{16}{3}a^2t$$

$$\sigma_{\max} = \frac{|M|}{W}$$

$$M = -40aF$$

$$= \frac{40aF}{\frac{16}{3}a^2t} = \frac{15}{2} \frac{F}{at}$$